

Nationality Detection Model

1) Project Overview

This project implements an image-based Nationality, Emotion, Age and Dress Colour Detection System.

The system combines Deep Learning and Computer Vision techniques to:

- Predict a person's nationality category from an uploaded image.
- Predict the person's emotion.
- Predict age, for Indian and USA nationality.
- Predict dress colour, for Indian and African nationality.
- Provide a graphical user interface (GUI) for image upload and result visualisation.

2) Problem Statement

The objective of this project is to develop a machine learning system capable of predicting a person's nationality, emotion, and, depending on the predicted nationality, their age and/or dress colour.

Decision Rule:

- Predicted nationality Indian: show age, dress colour, and emotion.
- Predicted nationality USA: show age and emotion.
- Predicted nationality African: show dress colour and emotion.
- Other nationality categories: show only nationality and emotion.

3) Dataset

UTKFace Dataset

Dataset Source:

Kaggle – UTKFace Dataset

Age, gender, and ethnicity labels are extracted directly from the image filenames. Ethnicity labels are remapped onto the four nationality buckets this task requires: White to USA, Black to African, Asian to Other, Indian to Indian, Others to Other. This is the dataset the nationality and age CNN is trained on.

FairFace Dataset

Dataset Source:

Kaggle – FairFace Dataset

A separately collected, more demographically balanced face dataset with its own race labels. It is never used during training – it is held out to test whether the nationality model generalises to faces it has never seen.

FER2013 Dataset

Dataset Source:

Kaggle – FER2013 Dataset

Emotion classes include:

- Angry
- Disgust
- Fear
- Happy
- Sad
- Surprise
- Neutral

Fashion Product Images Dataset

Dataset Source:

Kaggle – Fashion Product Images Dataset

Real clothing product photos labelled with actual colour names. Colour names for dress-colour prediction are built directly from this dataset by averaging pixel colours from real labelled photos per colour category, instead of hand-picking a fixed palette.

Dataset Structure:

Images and labels were organised per dataset – UTKFace encodes labels in the filename, FairFace and FER2013 provide a labelled CSV, and the Fashion dataset provides a styles.csv alongside its images.

4) Methodology

1. Data Preprocessing

- Face crops resized to 64 x 64 pixels (nationality/age model) or 48 x 48 grayscale (emotion model).
- Pixel values normalised to 0-1.
- Age targets scaled by /100 during training for stability.

2. Nationality and Age Classification Model

A Convolutional Neural Network with two output heads was trained.

Model Architecture:

- Convolution Layer – 32 filters
- Batch Normalisation + Max Pooling
- Convolution Layer – 64 filters
- Batch Normalisation + Max Pooling
- Convolution Layer – 128 filters
- Batch Normalisation + Max Pooling
- Dense Layer – 128
- Dropout – 0.3
- Softmax Output (nationality, 4 classes) + Linear Output (age)

3. Emotion Classification Model

A separate CNN was trained for 7-class emotion classification.

Model Architecture:

- Convolution Layer – 32 filters
- Batch Normalisation + Max Pooling
- Convolution Layer – 64 filters
- Batch Normalisation + Max Pooling
- Convolution Layer – 128 filters
- Batch Normalisation + Max Pooling
- Dense Layer – 256
- Dropout – 0.4
- Softmax Output Layer – 7 classes

4. Face Detection

OpenCV's frontal-face Haar cascade detects and crops the face region before either model sees the image.

5. Dress Colour Detection Logic

For each detected face:

- Crop the region below the face.
- Average the pixel colour in that region.
- Classify the colour into a main colour family by hue angle.
- Match to the closest labelled shade within that same family, built from the Fashion Product Images Dataset.

6. Final Decision Logic

The predicted nationality controls which extra fields (age, dress colour) are shown, per the Decision Rule described above.

5) GUI Development

Gradio was used to develop a graphical user interface.

Features:

- Upload a person's image
- Preview image
- Predict nationality
- Predict emotion
- Conditionally predict age
- Conditionally predict dress colour
- Display prediction output

6) Results

Nationality and Age Model

- Evaluated on a held-out UTKFace validation split, and separately on FairFace, a completely independent dataset never used in training.

Emotion Model

- Evaluated on a held-out FER2013 validation split.

Dress Colour Detection

- Evaluated on held-out Fashion Product Images Dataset samples, checking whether each sample is classified into the correct main colour family.

Exact accuracy figures vary slightly between runs (random train/validation split, random weight initialisation) – refer to the notebook's Accuracy Summary cell for the current run's numbers.

7) Visualisations

Included in notebook/repository:

- Training Accuracy Graph
- Validation Accuracy Graph
- Training Loss Graph
- Validation Loss Graph
- Confusion Matrix – Nationality
- Confusion Matrix – Emotion
- Age Prediction Scatter Plot
- Accuracy Summary Bar Chart

8) Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- Pandas
- Matplotlib
- Scikit-Learn
- Gradio

9) Future Improvements

- Train on the full datasets with more epochs.
- Benchmark a pretrained backbone (transfer learning) against the custom CNNs.
- Address FER2013's class imbalance with targeted data augmentation.
- Use a stronger clothing-segmentation approach for dress colour.
- Test on a wider variety of real-world photos.
- Deployment using Streamlit or Flask.

10) Conclusion

This project successfully combines Deep Learning and Computer Vision techniques to predict a person's nationality category, emotion, age, and dress colour from an uploaded image. The developed system trains two CNNs from scratch, builds a real-data-driven dress-colour reference table, evaluates performance across four datasets – including a genuinely independent test set, FairFace – and visualises results through an interactive Gradio GUI.